

Rabbit Hole

Deep Research Guide & 3-Week Team Execution Plan (3 people: **A** · **S** · **V**, 2–3 hrs/day each) — From Idea to Demonstration-Ready Prototype

End goal after 3 weeks: A live, demonstrable mobile web prototype showing the core loop — profile → hidden Fit Score → match suggestion → proximity date notification → tap/QR date logging → regional leaderboard with weekly reset — plus a validated research brief, a 3-min demo video, and a pitch deck.

Team Roles & Ownership

Person	Role	Owns
A	Product & Research Lead	User surveys & interviews, PRD, microcopy/content, seed data, QA & user testing, demo video, pitch deck, demo-day narration
S	Frontend Lead (Design → UI)	Figma wireframes & hi-fi design, all screens/UI, swipe & chat interfaces, animations, mobile polish, demo mode toggle
V	Backend & Systems Lead	Firebase (Auth/Firestore/Hosting), Fit Score engine, matching & proximity logic, date-logging (QR/double-confirm), leaderboard + weekly reset, push notifications, deployments

Rule: everyone commits to GitHub daily; A maintains **decisions.md**; daily 15-min standup (async on WhatsApp is fine — yesterday/today/blockers).

Part A — Deep Research: What to Investigate Before Building

1. Market & Competitor Research (Owner: A + S)

Area	Questions to Answer	How / Where
Competitors	How do Tinder, Bumble, Hinge, OkCupid, Iris Dating (AI face-based), Volar (fit-score dating), Teaser AI work? What are their engagement & monetization models?	Use the apps 20 min/day, read teardowns (ProductHunt, Medium, TechCrunch), Sensor Tower / data.ai public reports
Gap analysis	Which competitors already do gamification, leaderboards, or date verification? What is truly unsolved (ghosting, fatigue)?	Feature comparison matrix spreadsheet
Market size	Dating app market size (India + global), user demographics, ARPU, churn rates	Statista, Business of Apps, IBISWorld summaries
Failure post-mortems	Why did gamified dating apps fail/succeed? (e.g., Iris Dating shut down — why?)	Founder interviews, podcasts, post-mortem blogs

2. User Research (Owner: A — the most important Week-1 activity)

- **Surveys:** Google Form, 30+ respondents (18–30, current/past dating app users). Ask: biggest frustration, ghosting experience, willingness to pay, reaction to leaderboards & hidden scores.
- **Interviews:** 6–8 deep interviews (15 min each). Focus on: why they quit apps, what "progress" feels like, comfort with face scans and location tracking.
- **Key validation questions:**

- Will users accept a *hidden* score they can't see? (risk: feels unfair/opaque)
- Will they physically tap phones / scan a QR with a first date? (awkwardness risk)
- Is a weekly-reset leaderboard motivating or demoralizing?
- Would they pay? For what exactly?

3. Technical Feasibility Research (Owner: V, input from S)

Component	Research Tasks
Facial analysis	Options: Face++, AWS Rekognition, MediaPipe (free/on-device). Check accuracy, cost, consent needs. For demo: mock score, real ML is Phase 2.
Personality assessment	Big Five (OCEAN) short forms — free IPIP-10/20 questionnaires. Scoring = weighted sums.
Proximity detection	Web Geolocation API vs native GPS. Geofencing logic (Haversine distance). Permission UX.
Tap-to-log (NFC)	Web NFC is Android Chrome only ; iOS blocks it and NameDrop isn't open to third parties. Fallback: QR code scan or double-confirm — same UX, works everywhere.
Push notifications	Web Push API vs Firebase Cloud Messaging. Test deliverability.
Leaderboard & reset	DB query + Firestore scheduled function (weekly cron). Trivial.
Matching algorithm v0	Research score-band matching, Elo-style ratings (Tinder's old Elo). Start with weighted-sum, not ML.

4. Legal, Privacy & Ethics Research (Owner: V + A)

- **Biometric data:** face scanning falls under India's DPDP Act 2023 and GDPR/BIPA. Explicit opt-in consent flow required.
- **Location tracking:** proximity tracking is sensitive; check permission rules and foreground-only usage for demo.
- **Anti-discrimination risk:** hidden "attractiveness/personality score" + public leaderboard can attract backlash (Tinder Elo controversy). Research how Iris/Volar communicated AI scores publicly.
- **Abuse vectors:** fake date logs, score manipulation, stalking via leaderboard. Mitigations: region-level ranks only, no exact locations, anomaly flags.

Biggest risk to research first: public leaderboard + hidden face-based score = potential PR/ethical firestorm. Decide early: public names or anonymous rank-only? Validate with users in Week 1 (A owns this decision input).

5. Business Model Research (Owner: A)

- Benchmark pricing: Tinder Gold (~₹500/mo), Bumble Premium, HingeX. What do users actually pay for?
- Map monetization to features: event tickets (parties), score-band upgrades, unlimited matches, "fit hints".
- Research event-based dating businesses (Offline Club, Thursday) — events could be the differentiator.

Part B — 3-Week Day-wise Team Plan (2–3 hrs/day each)

Stack: Next.js + Tailwind (S) · Firebase Auth/Firestore/Hosting/Web Push (V) · Research/content/QA/deck (A). Research and build run in parallel.

Week 1 — Validate, Design & Architect

Day	A — Product/Research	S — Frontend/Design	V — Backend/Systems
1	Competitor teardown from user side (Tinder, Hinge, Iris, Volar — 20 min each). Draft survey (12–15 Qs) in Google Forms.	Competitor teardown from product side: feature matrix (gamification, matching, monetization). Shortlist tech stack for UI (Next.js vs React).	Research Fit Score algorithms (Elo, weighted-sum). NFC feasibility check → decide QR/double-confirm fallback. Start privacy checklist (DPDP/GDPR).
2	Finalize + distribute survey (WhatsApp/Instagram/college groups). Book interview slots.	Figma wireframes: signup, personality quiz, home/matches screens.	Research IPIP-10 personality quiz + scoring math. Draft Fit Score formula v0 (personality 40%, behavior 30%, date activity 30%).
3	Conduct 3 user interviews (15 min each). Log quotes.	Figma wireframes: chat, date-log, leaderboard, events screens.	Dev environment setup. Firebase test project + Next.js boilerplate spike (auth + one Firestore read/write working).
4	Conduct 3 more interviews. Start survey analysis.	Hi-fi mockups: design system (colors, typography, components) for 3 key screens.	Test Web Push + Geolocation API in browser. Build small QR-code prototype (scan → confirm).
5	Compile validation summary: survey data + interview insights + go/no-go on leaderboard anonymity model.	Hi-fi mockups: remaining screens. Clickable Figma prototype v1.	Design Firestore data model (users, matches, messages, dates, scoreEvents, leaderboards). Document it.
6	Write PRD: user flows, screens, score rules, ghosting penalties, monetization hypothesis. (30-min call with S+V for input.)	Review PRD → break frontend into component/task list on GitHub board.	Review PRD → define score-event schema + backend task list. Architecture diagram v1.
7	Recruit 5 beta testers for Week-2 user testing. Buffer/catch-up.	Finish clickable prototype. Buffer/catch-up.	Finalize architecture. Buffer/catch-up. Repo + branch strategy + GitHub project board ready.

Week 1 targets (high-end but achievable): 30+ survey responses · 6 interviews done · PRD signed off by all 3 · clickable Figma prototype of every screen · Firestore data model + Fit Score formula v0 documented · one live Firebase spike deployed. **If survey says users hate the hidden-score or public-leaderboard idea → adapt the model NOW, on paper, not in code.**

Week 2 — Build the Core Loop

Day	A — Product/Research	S — Frontend	V — Backend
8	Write ALL microcopy: quiz questions, notifications, empty states, error messages. Set up bug-tracker labels + daily QA log.	Project scaffold: Next.js + Tailwind, routing, app shell (bottom nav), design tokens from Figma.	Firebase prod project: Auth, Firestore, Hosting. CI/deploy pipeline. Empty app live on a public URL.

9	Create seed dataset part 1: 10 fake profiles (names, photos, bios, varied personalities).	Onboarding screens: signup form, photo upload UI, personality quiz UI (uses A's questions).	Auth flow wired end-to-end. Quiz scoring → Fit Score computed & stored as hidden Firestore field (never exposed in UI).
10	Seed dataset part 2: 10 more profiles. Test onboarding flow yourself, log every bug/UX friction.	Swipe screen UI (card stack, left/right gestures, animations).	Matching query: users within score band + city. Swipe → mutual right-swipe creates match document.
11	Bug triage: prioritize S/V fix list. Write demo script v0 (the exact story you'll show on Day 21).	Matches list + chat UI screens.	Realtime chat (Firestore listeners). Communication signal → score-event on messages/day.
12	Guerilla user testing: 3–5 beta testers use the half-built app, record reactions + drop-off points.	Date-log UI: proximity banner, QR screen, double-confirm screen, success/score-up animation.	Proximity engine: Geolocation + Haversine check between matched users (200m radius). QR/double-confirm logging → score boost for both.
13	Compile user-testing feedback → prioritized change list (max 5 items).	Implement top feedback fixes. Ghost/report flow UI (penalty trigger).	Score-event system: ONE update function wired to all signals (swipes, messages, date logged, ghost penalty).
14	Full E2E QA with 2 phones + 2 test accounts; document every issue found. Integration day.	Integration bugfixes (UI side).	Integration bugfixes (backend side). Deploy stable build.

Week 2 targets: Complete core loop works end-to-end on 2 real phones: signup → quiz → hidden score → swipe → match → chat → proximity banner → date logged via QR → both scores rise → ghost penalty fires. 20 seeded profiles live. 3–5 external users have touched the app and feedback is triaged. App deployed on a public URL by Day 14.

Week 3 — Gamification, Polish & Pitch

Day	A — Product/Research	S — Frontend	V — Backend
15	Pitch deck skeleton (10 slides). Problem + solution + research-findings slides (real user quotes from Week 1).	Leaderboard UI: regional ranks (rank + first name + city only), weekly-reset countdown timer.	Leaderboard backend query + Firestore scheduled function for weekly reset. Verify reset with a manual trigger.
16	Deck: market size, monetization, competition slides. Write demo-video voiceover script.	Events screen ("Rabbit Hole parties" list + RSVP). In-app notification center UI.	Web Push notifications: proximity alert + weekly reset alert. RSVP storage.
17	Finalize seed data in DB (scores spread so leaderboard looks alive). Record voiceover.	Demo Mode toggle UI (mock coordinates, skip real geolocation).	Demo Mode backend hooks. Seed-data script run in prod. Stable deploy.
18	Full deck draft complete (demo link, screenshots placeholders). Rehearse narration once, timed.	Polish pass: swipe/score-up animations, loading & empty states, mobile responsiveness on 3 screen sizes.	2-phone simultaneous stress test. Latency/perf fixes. Permission-denial fallbacks.
19	Record 3-min demo video (screen capture + voiceover). Backup for live-demo failure.	Team bug bash: fix UI bugs found.	Team bug bash: harden offline states, edge cases. Freeze feature set — bugs only from here.
20	Finalize deck: embed video link, user quotes, roadmap, team, ask. Print/share.	Final deploy. Capture clean screenshots for deck.	Final deploy + write tech-architecture slide content. Verify everything works on fresh devices/incognito.
21	Lead full rehearsal: 3-person demo choreography (who says what, who holds which phone). Buffer.	Buffer / last-mile polish.	Buffer / last-mile fixes.

Week 3 targets: Leaderboard with working weekly reset · notifications firing · events screen with RSVP · Demo Mode that works flawlessly without real GPS · 3-min demo video · 10-slide pitch deck with real research data · one full 3-person rehearsal done. **Demo-ready by Day 21, with Day 19 as the hard feature-freeze line.**

What to Deliberately SKIP in 3 Weeks (roadmap for later)

- Real facial-structure ML — mock it in demo, mark "Phase 2" in deck
- Real NFC tap (iOS blocks web NFC) — QR/double-confirm demonstrates identical UX
- Native iOS/Android apps — PWA installs to home screen
- Payments, real ML matching, background location tracking, anti-cheat systems — all roadmap items

Team Rules (all 21 days)

- 15-min daily standup (async WhatsApp ok): yesterday / today / blockers
- Everyone commits to GitHub daily — commit history is proof-of-work for the pitch
- A owns **decisions.md**: every design decision + rationale (becomes deck credibility)
- Disagreements get max 10 minutes, then the domain owner decides (A=product, S=UI, V=backend)
- If someone misses a day: their buffer day (7/14/21) absorbs it — don't cascade delays